Comparative Analysis of S&P 500 and NASDAQ Indices: A Machine Learning Approach to Understanding Differential Market Sensitivities and Growth Stock Dynamics

Leyao Wang *

College of Business Administration, Capital University of Economics and Business, Beijing, 100070, China

* Corresponding Author Email: 32021020190@cueb.edu.cn

Abstract. This paper delves into the implementation of a Walking Forward Machine Learning Model designed for the construction and forecasting of portfolios comprised of randomly selected stocks from the S&P 500 and NASDAQ indices. The primary objective of the study is to scrutinize the attributes of technology stocks, specifically assessing whether they exhibit characteristics typical of growth stocks. Furthermore, the research aims to discern potential disparities in predictive characteristics between technology stocks and standard equities. The passage elucidates the research methodology, encompassing data collection, model establishment, and evaluation processes, along with insights into the intricacies of portfolio construction. The empirical findings shed light on the model's efficacy in predicting stock returns and its aptitude in crafting investment portfolios. The concluding remarks underscore the significance of adjusting the model's learning rate in accordance with the distinct traits of various markets, emphasizing the need for a nuanced approach to enhance its overall generalization performance. This study contributes valuable insights into optimizing machine learning models for effective portfolio management in dynamic financial markets.

Keywords: Stock Indices, Machine Learning, Portfolio Management.

1. Introduction

Stock market prediction poses a significant challenge in the realm of time series forecasting due to the inherent characteristics of the stock market as a nonlinear, dynamic, noisy, and chaotic system [1]. The price of stocks is influenced by various factors, including political events, company policies and news, economic conditions, interest rates, and investor sentiments [2]. In the field of financial engineering, accurately predicting time series data is a crucial and complex task. It requires addressing trends, seasonal variations, and the interdependencies and correlations within the data. Traditional validation methods such as Leave-One-Out Cross-Validation and k-fold Cross-Validation may not be suitable for time series data as they tend to overlook the continuity and dependency inherent in such data, leading to overfitting or underfitting of models. To overcome this challenge, this research aims to adopt validation methods that are better suited to the characteristics of time series data. Rolling Time Window Cross-Validation is an effective approach that takes into account the sequential and dependent nature of time series [3]. In this method, training sets are continuously adjusted using moving time windows, ensuring that models are always based on the most up-to-date data.

Conventional theories suggest that technology stocks, known as growth stocks, are highly sensitive to market fluctuations and technological innovations. This implies that they may perform well during market booms but suffer during downturns or economic recessions. The objective of this study is to utilize a Walking Forward Machine Learning Model to construct and forecast portfolios consisting of randomly selected stocks from the S&P 500 and NASDAQ indices. Machine learning models, including decision trees, regression analysis, and Bayesian networks, have become powerful tools for predicting future events and optimizing outcomes [4]. By shifting the focus from parameter significance to model structure and dynamic characteristics, this research enriches the field of financial and economic theory [5]. Through a comparison of predicted and actual results, the study aims to analyze whether technology stocks exhibit the characteristics of growth stocks. Furthermore, it aims to explore potential differences in prediction between technology stocks and standard stocks (S&P 500), with the hope of uncovering insights that will benefit future research.

Analysis of Key Moments: This research will select three significant moments that have had a profound impact on technology stocks: the Facebook lawsuit in December 2018, the public registration opening of OpenAI in November 2022, and the disclosure by the White House in May 2023 regarding potential further regulation targeting artificial intelligence products. The objective is to investigate how these pivotal events influenced the price movements of technology stocks and evaluate the effectiveness of the Walking Forward Model in predicting the outcomes of these black swan events. Black Swan events are rare and seemingly random in nature. In the influential paper by Nassim Nicholas Taleb, it is argued that Black Swan events cannot be reliably predicted, and instead, it is essential to be prepared for them at all times [6].

2. Methodology

2.1. Data Collection

This code documentation provides an overview of the data collection, downloading, and processing process, laying the foundation for subsequent model construction and portfolio building.

In this section, the code imports 50 random samples from both the SPY index and the Nasdaq 100 index. First, the code sets a random seed using `np. random. Seed (5)` for result reproducibility. Then, the code reads the list of tickers from a Wikipedia page and retrieves a random sample of tickers based on the sample index, selecting 50 random samples from the S&P 500 index. The code defines a function called `get_nasdaq_tickers () ` to fetch the stock tickers of the Nasdaq 100 index from a Wikipedia page. Then, the code randomly selects 50 stock tickers from the Nasdaq 100 index and stores them in `nasdaq_list`.

This section downloads stock price data. The code uses the `yf.download` function to download the stock price data for the selected symbols, namely the S&P 500 index and the Nasdaq 100 index, from Yahoo Finance. The downloaded data is stored in `prices_df` and `prices_df_nas`. The data is downloaded within the range from `start_date` to `end_date`. Then, the code uses the `dropna` function to remove any columns with missing values. This section performs data cleaning. It uses the `dropna` function to remove any remaining missing values from the S&P 500 index and the Nasdaq 100 index data. `axis=1` indicates that columns with missing values are removed, while `axis=0` indicates that rows with missing values are removed. `how='all'` indicates that rows or columns consisting entirely of missing values are removed.

This section calculates returns. The code uses the `pct_change` function to compute the returns of the S&P 500 index and the Nasdaq 100 index. `pct_change (20) ` calculates the returns over a 20-day period, while `pct_change (1) ` calculates the daily returns. Then, the code uses the `resample` function to resample the return data to a 30-day frequency. This section plots boxplots. The code defines a function called `remove_outliers` to remove outliers from the return data. Then, the code uses `plt.boxplot` to plot boxplots of the returns of the S&P 500 index and the Nasdaq 100 index, visualizing the distribution of returns.

This section prints the return data of the S&P 500 index and the Nasdaq 100 index. This data report provides a clear overview of the data collection, downloading, and processing process, laying the foundation for subsequent model construction and portfolio building. While the author successfully downloaded 50 randomly selected stocks, there may be cases where certain stocks were not included due to their listing period not being covered or being suspended or delisted within the current time range.

2.2. Modelling

The establishment and optimization of the model is a key step in this study. The author adopted the Walking Forward Machine Learning method and used Ridge regression model for prediction. The following will detail the process of model establishment and the process of model hyperparameter optimization and evaluation.

2.2.1. Model Establishment

The author first defines a time sliding window method to control the size of the training set. By setting variable sliding window sizes, the research can adjust the minimum time period of the training set to ensure that the model has sufficient data for training. Next, the author selected a set of hyperparameters (alpha_values) as tuning parameters for the Ridge regression model. These hyperparameters will be adjusted in the subsequent model optimization process. The research uses the train_and_evaluate_model function to implement model training and evaluation. The function accepts the data of training set and testing set as inputs, and trains the model by iterating over different alpha values. In each time sliding window, research take one stock as the target variable (y) and use the returns of other stocks as feature variables (x) to train the model [7]. By adjusting the hyperparameter alpha, the author obtained the optimal model with the smallest mean squared error (MSE).

2.2.2. Hyperparameter Optimization

The author optimizes the model by iterating over different training set sizes (train_sizes). For each training set size, the research uses the train_and_evaluate_model function to train the model, and select the optimal alpha value with the smallest MSE. For each training set size, the code calculates the predicted results of the model, and calculate the MSE and mean absolute error (MAE) of the prediction results. By comparing the MSEs under different alpha values, the author selects the optimal alpha value with the smallest MSE as the hyperparameter of the model.

2.2.3. Model Evaluation

The author uses MSE as an evaluation metric for model performance. MSE measures the average squared error between the predicted value and actual value of the model, which can reflect the accuracy and precision of the model. In addition, the research can use other evaluation indicators such as MAE to further evaluate the performance of the model. MAE measures the average absolute error between the predicted value and actual value of the model, which can provide a more comprehensive evaluation of model errors.

Through the above modeling and optimization process, the author obtained a predictive model based on S&P500 data, and selected the optimal model through hyperparameter tuning and evaluation metrics. This model can be used to predict stock returns and provide important references for investment decisions.

2.2.4. Portfolio Construction

Based on the predictive results of the model, the author can construct investment portfolios. The specific steps include the following:

(1) Using CVXPY library for portfolio optimization: The author utilizes predicted stock returns and covariance matrix to optimize investment portfolios. By minimizing the volatility of portfolios and satisfying some constraint conditions such as weight sum of 1 and non-negative weights, the research can obtain the optimal weight for investment portfolios.

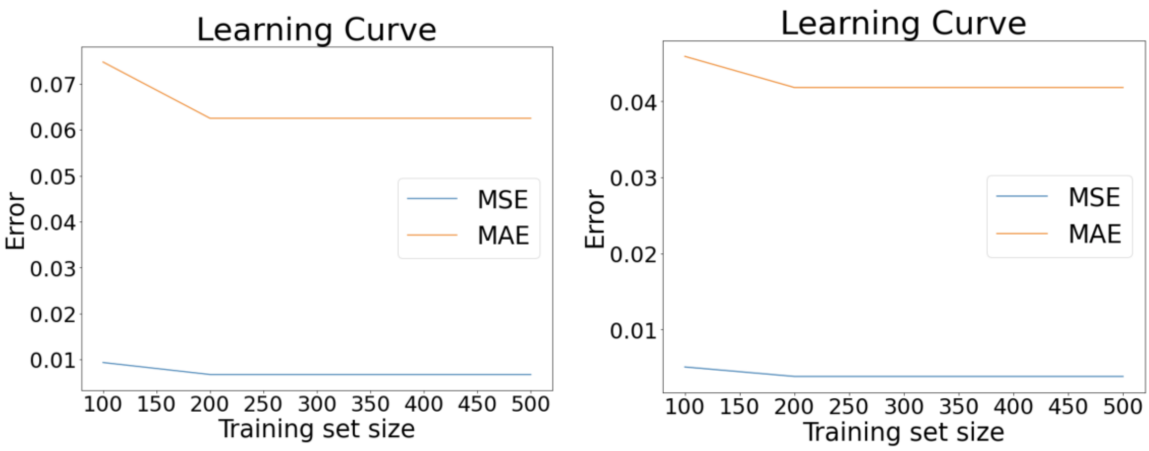
(2) Portfolio backtesting: Apply each time window's portfolio weight to each stock's daily return rate to calculate the cumulative return of the portfolio. At the same time, research also compare it with the cumulative return of a benchmark index (such as SPY) to evaluate the performance of the portfolio. Through the above investment portfolio construction process, the code can make effective investment decisions based on the predictive results of the model and evaluate the return of investment portfolios.

3. Empirical Results

3.1. MSE & MAE in the Process Modelling Evaluation

In the process of modeling the S&P 500 index, an analysis of Mean Squared Error (MSE) and Mean Absolute Error (MAE) reveals interesting insights (see Figure 1(a)). The research observes that beyond 200 samples, the reduction in MAE becomes marginal, indicating that additional data provides limited benefits in decreasing the mean absolute error. This suggests that the model's predictive accuracy plateaus after a certain sample size. Furthermore, the stability of MSE across the sample range implies that the model might be sensitive to certain patterns or noise within the data. It is also possible that the training data lacks sufficient information to further reduce the error. Future research should focus on strategies to mitigate the impact of noise on the model's performance.

Based on these findings, it appears that the current model has reached its threshold in terms of predictive accuracy. This indicates a need to explore model improvements or alternative approaches to enhance the accuracy of future predictions. Interestingly, this phenomenon also applies to return forecasts for the NASDAQ index (see Figure 1(b)). This suggests that, for this particular method, a sample size of 200 may be a limit in terms of achieving optimal accuracy. To increase the accuracy further, it recommends expanding the time window for data collection and analysis.



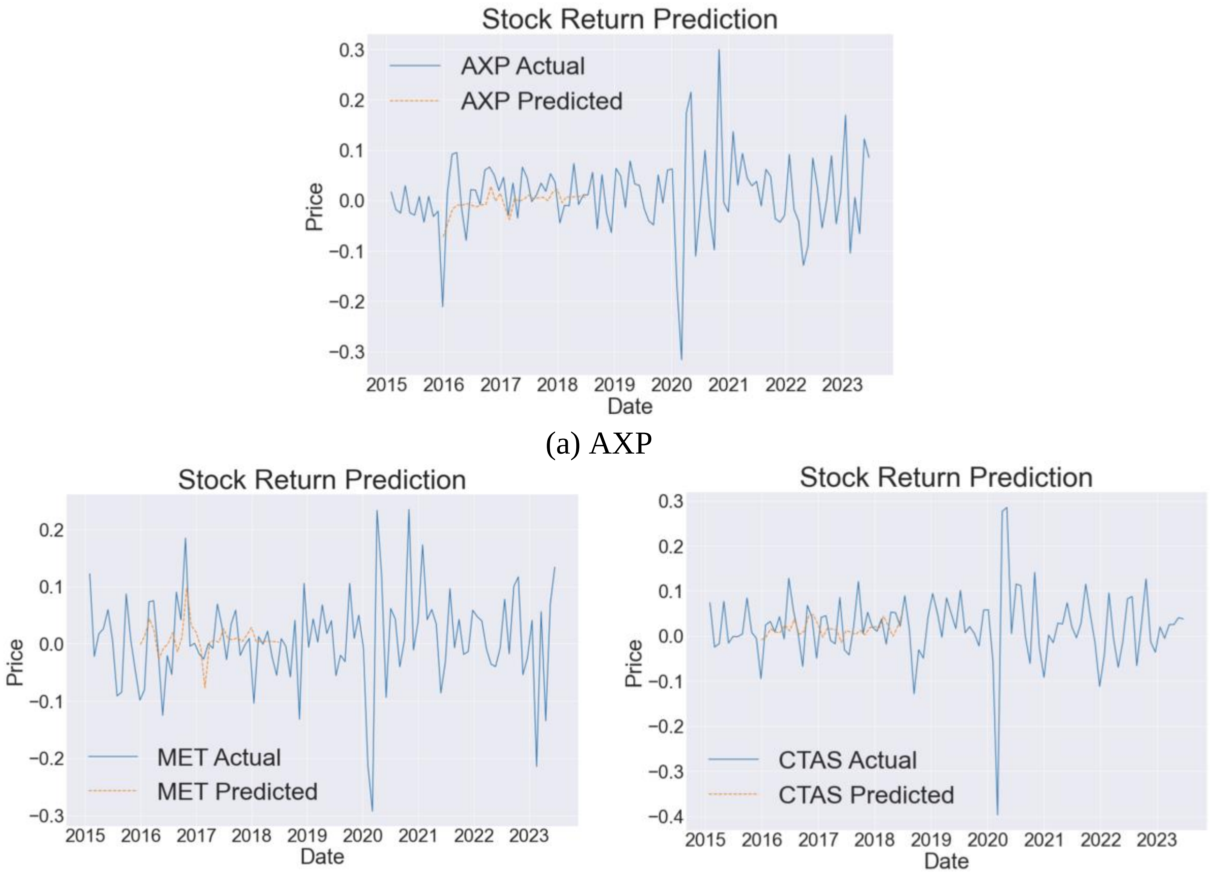
(a) S&P500 (b) NASDAQ

Figure 1. The MSE&MAE learning curve (Photo/Picture credit: Original)

Simultaneously, through the optimization of MAE and MSE, the research has derived different ridge alpha values for the two indices: 0.01 for the S&P 500 and 1 for NASDAQ. The alpha parameter, also known as the regularization parameter, controls the degree of constraint on the model's weights. Different alpha values affect how closely the model fits the data and its ability to generalize to new data. Based on the results, the author can speculate that NASDAQ stocks, which are more heavily weighted towards technology stocks with potentially greater price volatility, require a larger alpha value. On the other hand, the S&P 500, which includes a broader range of industries, may be relatively more stable and thus require a smaller alpha value. This finding indirectly supports the theory mentioned at the beginning of the paper.

3.2. S&P 500 and NASDAQ Prediction vs. Returns Display

To visually analyze the performance of the model, the author has developed a code to dynamically display a comparative trend graph of actual versus predicted results for a randomly selected stock from the pool of 50. This graph presents the actual and predicted returns of a particular stock over time.



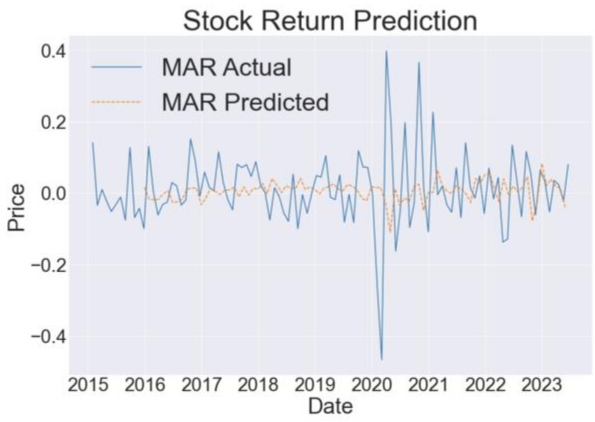
(a) AXP

(b) MET (c) CTAS

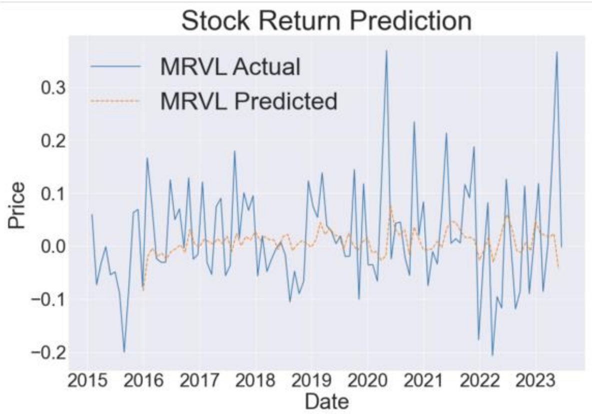
Figure 2. Stock return forecast in SPY Index (Photo/Picture credit: Original)

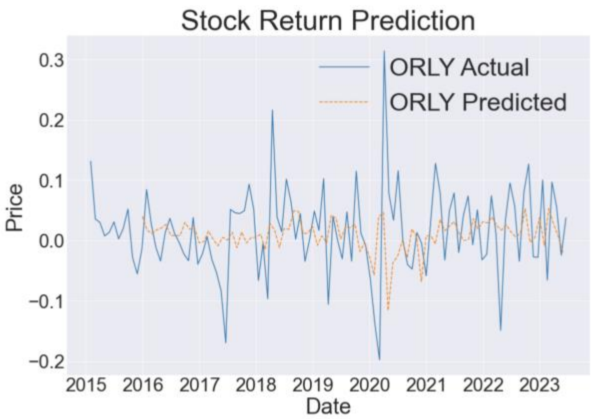
Figure 2 and Figure 3 illustrate the fluctuation of the stock's return rates over time, with both the actual and predicted lines showing significant volatility. There are periods, such as the beginning of 2015 and mid-2020, where the prediction closely follows the actual trend. However, in other periods, notably the start of 2020 and 2022, there are larger discrepancies, indicating the model's inability to accurately predict the stock's performance during these times.

In comparing the predictive outcomes against the SPY index with those against the NASDAQ, it becomes evident that predictions for the S&P 500 are more accurate. Observationally, the Nasdaq exhibits significantly greater volatility, a factor the model did not anticipate. This highlights the need for future improvements in the author's forecasting approach for high-volatility data, such as the application of differencing methods. Predictions seem to align with actual data in certain periods. However, there are intervals, especially in early 2020 and early 2022, where the prediction significantly deviates from reality, suggesting that the model failed to accurately foresee the stock's behavior during volatile periods both in SPY and Nasdaq100 Indices. A sharp decline in actual return rates is observable at the beginning of 2020. The predictive line failed to capture this downturn, possibly due to the model not accounting for the impact of such rare events. To improve the model's predictive power, it may be necessary to consider more complex time-series models, such as ARIMA or GARCH, or to incorporate macroeconomic indicators as predictive variables. Furthermore, it's essential to consider whether the model includes mechanisms to adapt to extreme market events and whether special modeling for such events is required [8].



(a) MAR

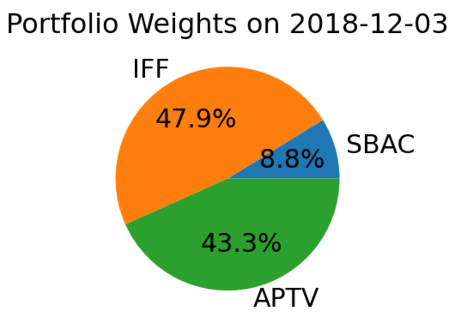




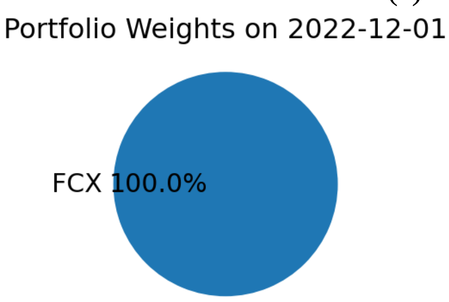
(b) MRVL (c) ORLY

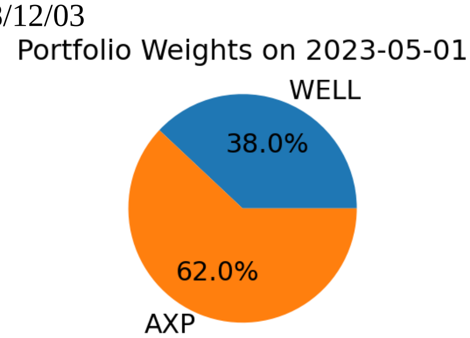
Figure 3. Stock return forecast in NASDAQ Index (Photo/Picture credit: Original)

3.3. S&P500 & NASDAQ Portfolio Weights Comparison



(a) 2018/12/03

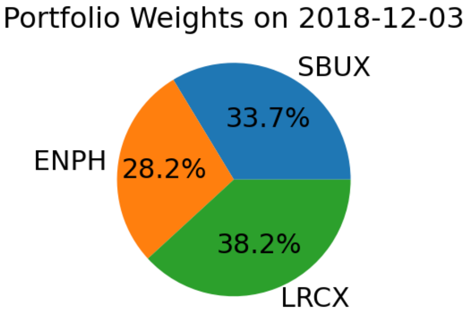




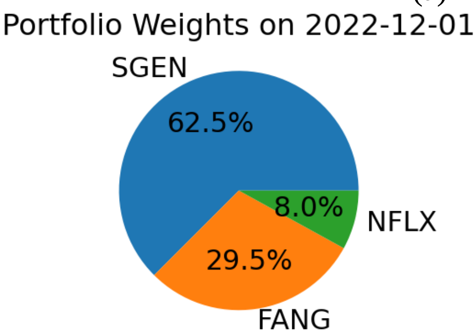
(b) 2022/12/01 (c) 2023/05/01

Figure 4. S&P500 weights on 3 time points (Photo/Picture credit: Original)

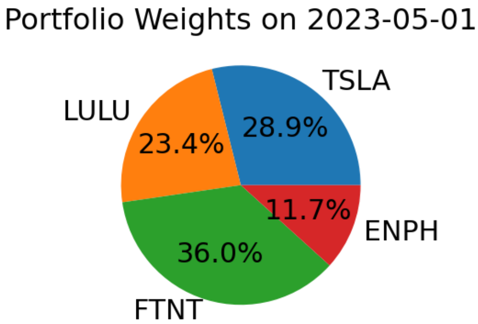
SPY index portfolio (see Figure 4): Despite the selection of three tech-related milestones, the S&P 500's top ten constituents are tech-centric firms such as Apple, Microsoft, Amazon, Tesla, Alphabet A and C shares, Berkshire Hathaway B, United Health, Nvidia, and Johnson & Johnson [9]. Yet, none of these tech-linked stocks were included in the portfolios at these specific times. The sole tech-affiliated stock chosen in May 2023 was WELL, a company dedicated to the digitalization of healthcare. The debut of OpenAI didn't result in a direct inclusion of Microsoft in the portfolio. This could be due to covariance constraints designed to minimize risk rather than increase the weight of tech stocks. During this period, a mining company was allocated instead, a decision that requires further analysis.



(a) 2018/12/03



(b) 2022/12/01



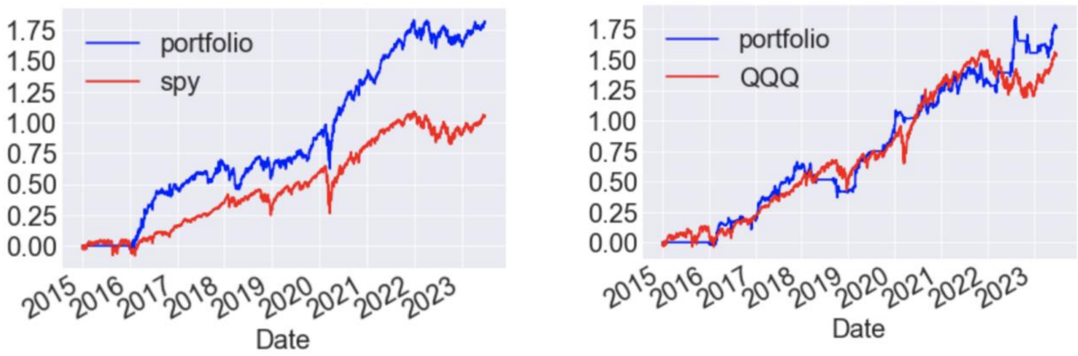
(c) 2023/05/01

Figure 5. NASDAQ weights on 3 time points (Photo/Picture credit: Original)

NASDAQ index portfolio (see Figure 5): At the time of OpenAI's release, the portfolio opted for an 8% stake in Netflix, an energy company involved in hydrocarbon extraction, and a biotech company, without selecting companies directly associated with OpenAI's supply chain. This indicates that the event did not significantly influence the optimal portfolio outcome [10]. Additionally, the forecasting failed to predict the severe volatility trend in tech stocks and the impact of market fluctuations, leading to a discrepancy between the projected and actualized portfolio returns. Across all three time points, 50% of the portfolio composition was unrelated to OpenAI, despite the inclusion of Tesla. Overall, the portfolio tended toward companies reliant on the tangible economy, such as e-commerce retail, energy development, and new energy vehicles, with fewer internet technology firms, notably Netflix and a cybersecurity company.

3.4. Portfolio Compared to Benchmark

Figure 6 indicates that the portfolios of both indices have surpassed their benchmark indexes, with the S&P 500 aligned with the SPY ETF and the NASDAQ with the QQQ ETF. However, the author harbor doubts about the accuracy of these predictions in practice. This is due in part to questions about whether the forecasted results can genuinely align with actual prices, and also because the simplicity of the model's factors may not account for market risks, potentially resulting in forecast biases.



(a) S&P500 (b) NASDAQ

Figure 6. Portfolio return Benchmarking (Photo/Picture credit: Original)

4. Conclusion

This study established a machine learning model based on S&P500 data to predict stock returns and constructed an investment portfolio based on the predicted results of the model. Through optimizing hyperparameters and evaluation metrics, the author obtained effective predictive models and optimized investment portfolios. This is important for improving investors' decision-making capabilities and achieving investment goals. Future research can consider incorporating more feature variables and optimization methods to improve model prediction accuracy.

After studying the sensitivity of machine learning models in different markets, the research has found that market volatility impacts the learning rate of the model. Hence, for the same model, while holding other variables constant, the research can cater to the needs of different markets by training with the optimal alpha values. This indicates that to ensure the model's generalization performance across various markets, a dynamic approach to the model's learning rate is necessary. In practice, the research can select the appropriate alpha values based on the characteristics of different markets to improve the adaptability and generalization performance of the model.

By delving into the significance and optimal range of the alpha value in machine learning, the author conclude that it is essential to dynamically adjust the model's learning rate when facing different markets to enhance its generalization performance in diverse conditions. Therefore, the research recommends that during the training process of machine learning models, the learning rate alpha should be dynamically adjusted according to the specific characteristics of different markets to ensure the model's generalizability in its applications.

References

[1] Deboeck, G. J. (1994). Trading on the edge: Neural, genetic, and fuzzy systems for chaotic financial markets. New York: Wiley.

[2] Wang, J. Z., Wang, J. J., Zhang, Z. G., & Guo, S. P. (2011). Forecasting stock indices with back propagation neural network. Expert Systems with Applications, 38, 14346 - 14355.

[3] Ren, J. (2023). Financial technology stock index prediction research based on LSTM combination network transfer model, Shandong University.

[4] Hu, Y. (2022). Research on the importance of investment factors in U.S. stocks based on machine learning, University of International Business and Economics.

[5] Su, Z., Lu, M., & Li, D. X. (2017). Financial empirical application of deep learning: Dynamics, contributions, and prospects. Financial Research, (05), 111 - 126.

[6] Musgrave, G. (2009). The Black Swan: The impact of the Highly improbable. In Business Economics, 44, 123 - 125.

[7] Pan, L., & Xu, J. G. (2011). Risk and characteristic factors of A-share market. Financial Research, (10), 140 - 154.

[8] Kuang, Y., & Liang, Z. J. (2012). Prediction analysis of the S&P 500 index based on ARIMA model. Modern Business and Trade Industry, 24 (14), 100 - 102.

[9] Akter, N., & Nobi, A. (2018). Investigation of the Financial Stability of S&P 500 Using Realized Volatility and Stock Returns Distribution. Journal of Risk and Financial Management, 11 (2).

[10] Huang, N. J., & Yu, M. Z. (2018). Research progress on the impact of machine learning on economics studies. Economic Dynamics, (07), 115 - 129.